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SUPERSEDING
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MILITARY SPECIFICATION

SONAR TEST SET AN/BQM-1

1. SCOPE

1.1 This specification covers the Sonar Test Set AN/BQM-1 suitable for determining the operating performance of listening sonar equipments.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-1 - Electron Tubes and Crystal Rectifiers.
MIL-T-27 - Transformers and Inductors (Audio, Power and Pulse).
MIL-T-945 - Test Equipment, for Use with Electronic Equipment General Specification.
MIL-C-3432 - Cable, Flexible, and Extra Flexible 300 and 600 Volts.
MIL-C-5015 - Connectors, Electric, "AN" Type.
MIL-M-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized.
MIL-E-15090 - Enamel, Equipment, Light Gray (Formula No. 11.
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.

FSC 6625

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MILITARY (cont'd.)

MIL-M-16616 - Manual, Technical, Electronic Equipment.
MIL-I-16910 - Interference Measurement, Radio, Methods and Limits, 14 Kilocycles to 1000 Megacycles.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-200 - Electron Tubes, Selection and Use of and Navy Supplement thereto.
MIL-STD-242 - Electronic Equipment Parts Selected Standards.
MIL-STD-415 - Test Points and Test Facilities for Electronic Systems and Associated Equipment Design Standard for.
MIL-STD-758 - Packaging Procedures for Submarine Repair Parts. Utilizing Transparent Flexible Heat Sealable Film.
MS3106 - Connector, Plug, Electric, Straight.
MS24663 - Connector, Plug, Electrical - With Spring Loaded Pivoted Grounding Blade, Polarized.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St. New York 16, N. Y.)

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3. REQUIREMENTS

3.1 Preproduction model.- One preproduction model shall be furnished for the tests specified in 4.3.

3.2 Reliability requirements.-

3.2.1 Reliability.- Reliability of operation shall be stressed throughout the design and manufacture of the equipment. The equipment will be subjected to continuous use for long periods under the varied and severe conditions of military service without overhaul and with little maintenance. To meet these conditions, it is imperative that reliability of operation be considered of prime importance in the design and manufacture of the equipment. The manufacturer shall employ all methods possible in the process of manufacture which will assure quality and maximum reliability consistent with the state of the art. In the functional application of parts to the equipment circuits, adequate factors of safety shall be provided by suitable deratings from the part specification values in order to insure high equipment reliability under all service conditions. The design shall include all possible features which will result in reliable and stable operation with reduced requirements for maintenance, and simplified maintenance, thus reducing requirements for highly skilled maintenance personnel.

3.2.2 Service conditions.- The equipment shall be designed to give optimum performance and reliable service during continuous or intermittent operating periods of at least 200 hours under the conditions specified herein without the necessity for major adjustments or servicing.

3.3 General requirements.- Unless otherwise specified, this equipment shall meet the requirements of MIL-E-16400.

3.3.1 Material.- The material used in this equipment shall be selected in accordance with the fungus-inert materials requirement of MIL-E-16400. The general fungus-resistant treatment and the requirement for coating soldered joints and wire with varnish, as specified in MIL-T-945, shall not apply.

3.3.2 Selection of parts.- Unless otherwise specified herein, parts used in this equipment shall be selected in accordance with MIL-E-16400 from the listings of MIL-STD-242. Requests for approval of nonpreferred parts to be used where there are no applicable preferred parts shall be submitted not less than 120 days before the delivery of any preproduction, or production model.

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3.3.2.1 Parts of the equipment not covered by specifications subsidiary to MIL-E-16400 shall be capable of withstanding the fungi test specified in 4.5.1.

3.3.2.2 Electron tubes.- Electron tubes shall be selected in accordance with MIL-STD-200 and the Navy supplement thereto. If it is determined that none of the tube types listed in MIL-STD-200 permit conformance with the other requirements of this specification, the contractor shall immediately submit the following information to the procuring agency for approval:

(a) The type designation of the tube the contractor proposes to use.
(b) Details of critical parameters.
(c) Schematic diagrams showing applicable voltages, currents, and impedances.
(d) A statement by the manufacturer of the proposed tube that MIL-E-1 specification sheets will be prepared and that the proposed tube will be manufactured and inspected in accordance therewith if the use of the proposed tube is approved by the procuring agency.

3.3.2.3 Transformers and inductors.- All audio, power, and pulse transformers and inductors shall conform to grade 4, class R, S or T, life expectancy X of MIL-T-27.

3.3.2.3.1 Transformer diagrams.- A transformer diagram shall be printed on the case of each transformer, or on a plate attached to the case. The diagram shall also be available in the technical manual.

3.3.2.3.2 Transformer data.- All data pertinent to the transformer type shall be marked on the case as indicated in MIL-T-27.

3.3.3 Temperature and humidity.- This equipment shall operate within the accuracies specified herein over the entire temperature range from 0° to plus 50°C. with a relative humidity of 95 percent and shall conform to class 4 of MIL-E-16400, except as specified herein.

3.4 Functional description.- This equipment shall provide noise signals of known levels for use in determining the performance of listening sonar equipments and capabilities for measuring the levels of external signals. Capabilities to provide external signals in the proper form and levels for sonar testing shall be included.

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3.5 Composition.- This equipment shall consist of the following items and parts specified herein and as required to make up a complete equipment:

(a) Noise Generator (see 3.6.2).
(b) Test Target Signal Amplifier (see 3.6.3).
(c) Alignment Signal Amplifier (see 3.6.4).
(d) Function Selector Circuit (see 3.6.5).
(e) Meter Circuit (see 3.6.6).
(f) Front Panel (see 3.8.7).
(g) Junction Box (see 3.6.12).
(h) Accessories (see 3.12).
(i) Power Supply (see 3.6.11).

3.6 Performance requirements.- The equipment shall conform to the requirements specified hereinafter. Figure 1 shows a diagram of an equipment having performance characteristics similar to those specified herein and should be used only as a guide.

3.6.1 Warmup.- The equipment shall be fully operable within 5 minutes after the main power is turned on.

3.6.2 Noise generator.- The noise generator shall have an output that is flat within plus or minus 1 decibel (db) from 1 to 40 kilocycles (kc.) The output level at the frequencies less than 1 kc. and above 40 kc. shall not rise above the level in the 1 to 40 kc. region. (A type 6D4 tube with a magnetic yoke may be used to fulfill this requirement.)

3.6.3 Test target signal amplifier.- The test target signal amplifier shall have an output characteristic with a slope of minus 12 db + 3 db - 0 db per octave from 3 to 40 kc. when working into the test target hydrophone shown on figure 1. Responses above 40 kc. shall not exceed the level at 40 kc. and responses below 1 kc. shall have at least a slope of 6 db per octave. For proper operation, the required r.m.s. noise voltage shall be approximately 4.9 volts.

3.6.3.1 With a sinusoidal input the test target amplifier shall be capable of delivering an output of at least 5 volts root mean square (r.m.s.) to the test target hydrophone with less than 5 percent distortion at the frequency of maximum response.

3.6.3.2 The test target ON-OFF switch shall be spring loaded with the normal position OFF.

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3.6.4 Alignment signal amplifier. - The alignment signal amplifier voltage versus frequency characteristic shall be a smooth curve which is symmetrical about 2 kc., 1 db down at 5 and 0.8 kc., 3 db down at 10 and 0.4 kc., 8 db down at 20 and 0.2 kc., 15 db down at 40 kc., and 27 db down at 80 kc. The following tolerances shall apply to the above characteristics:

Range of frequency Kc.	Tolerance (plus or minus) Db
0.2 to 20	2
20 to 40	3
40 to 80	4

3.6.4.1 Amplifier output shall be push-pull with a center connection each side having a nominal output resitance of 1 ohm (see 3.6.4.4 and figure 1). Output voltage shall be controlled by means of a 10 ± 0.5 db per step attenuator over the range 40 db below 1 volt to 120 db below 1 volt. Attenuator input and output impedance of each side shall be 500 ohms. Cumulative error in the attenuator shall not exceed 3 db (see suggested circuit shown on figure 2).

3.6.4.2 With a 2 kc. sinusoidal input, the alignment signal amplifier shall be capable of delivering to the attenuator a signal of 20 volts r.m.s. with less than 5 percent distortion.

3.6.4.3 Provision shall be made for injecting sinusoidal voltages from an audio oscillator into the alignment signal circuits (see 3.6.4.4 and figure 1) in such a manner that they will be controlled by the 10 db per step attenuator in the same manner and over the same range as the internally generated noise voltages. The voltage versus frequency characteristic of the circuits to which the sinusoidal voltages are applied shall be flat within plus or minus 1 db from 500 cycles per second (c.p.s.) to 30 kc.

3.6.4.4 A junction box shall be provided which contains the push-pull circuit for injecting alignment signals into the sonar set with which the subject equipment is used (see figure 1). The phasing of the alignment signal shall be controlled by an RLI test switch specified in 3.6.5. The junction box will be permanently located at the preamplifier input terminals of the listening sonar.

3.6.5 Function selector circuit. - An RLI test switch shall be provided with five settings; operate, L (left) C (center), R (right), and equal gain. This switch shall change the time relationship of the output in such a manner as to simulate both "on target" and "off target" positions. A suggested circuit is shown on Figure 1. The output transformer shall work into a 125-ohm load.

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3.6.5.1 With the RLI test switch in the "C" position, the output voltages shall be equal within 0.2 db and 180 degrees out of phase.

3.6.5.2 With the RLI test switch in the "L" or "R" position, the output voltages shall be equal within 0.3 db, one signal having a 2.27 ± 0.03 microsecond (ms.) delay in relation to the other in the frequency range of 500 c.p.s. to 30 kc. under normal operating conditions.

3.6.5.3 With the RLI test swtich in the "Equal Gain" position, the output voltage shall be delivered to only one of the output resistors.

3.6.5.4 With the selector switch in the "Operate" position, no voltage shall be supplied to the alignment circuit.

3.6.6 Meter circuit. - A meter shall be provided, as shown on figure 2, to be used for the following purposes:

(a) As an alternating current (a.c.) voltmeter for monitoring the voltage across the test target hydrophone.
(b) As a calibration and interpolation meter for the alignment signal.
(c) As an a.c. voltmeter with a minimum impedance of 15,000 ohms per volt for measuring external voltages over the range of 0 to 10 volts r.m.s. accurate within plus or minus 5 percent at full scale.

3.6.6.1 The functions specified in 3.6.6 (a) and (b) shall be accomplished by using the switching specified in 3.6.5.1 through 3.6.5.4 and schematically shown on Figure 1, and by engraving two scales on the meter face as follows:

Scale A: 0 to 10 volts in 0.5 volt increments.

Scale B: (a) A calibrate point at which the 10 db per step attenuator correctly indicates the alignment signal output within plus or minus 1 db when the needle is lined up with it.
(b) A decibel scale graduated 0 to -10 db in 1 db steps in which the "0" point coincides with the calibrate point.

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3.6.7 Signal selector switch. - A signal selector switch shall be provided as a means of choosing between noise and external oscillator.

3.6.8 Variations with temperature. -

3.6.8.1 The output level shall vary not more than 3 db at any frequency over the temperature range of 0° to 50° Centigrade (C.).

3.6.8.2 The time delay in the "R" and "L" positions of the RLI test switch shall vary not more than plus or minus 10 percent from 2.27 ms. over the temperature range of 0° to 50°C.

3.6.9 Signal inputs. - The following signal inputs shall be provided to the equipment.

3.6.9.1 External oscillator. - The external oscillator input shall be through a telephone type jack which automatically disconnects the alignment amplifier from the meter and alignment output circuits when a plug is inserted and connects the external signal to the meter and alignment output circuits.

3.6.9.2 Voltmeter input. - The voltmeter input shall be through a telephone type jack which automatically disconnects the meter circuit from internal signals when a plug is inserted and connects the external voltage to the meter circuit.

3.6.10 Signal outputs. - The following signal outputs shall be provided.

3.6.10.1 Alignment signal. - The alignment signal output shall be through a terminal strip for direct cabling to the junction box.

3.6.10.2 Test target signal. - The test target signal output shall be through a terminal strip for direct cabling to the test hydrophone.

3.6.10.3 Noise signal. - The noise signal output for the AN/BQR-2B simulated target shall be through a telephone type jack which automatically disconnects the test target hydrophone, bypasses the test target switch, and connects the output of the test target amplifier to the output when a plug is inserted.

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3.6.11 Power supply.-

3.6.11.1 Regulation.- The degree of regulation of the power supply shall be sufficient to allow the equipment to function within the requirements of this specification and to insure that no noticeable changes in the output signals occur when the power line input voltage and frequency, including transients, are within the range of 3.7.1.

3.6.11.2 Ripple.- The a.c. ripple components of direct current (d.c.) voltages shall be of such low magnitude that no undesirable beating or unstable signals shall occur in the output.

3.6.11.3 Ground.- The design of the power supply shall permit optimum performance of the equipment when either or neither side of the power supply line is grounded.

3.6.11.4 Transformer hum.- There shall be no audible transformer hum when the equipment is in operation.

3.6.12 Junction box.- A junction box suitable for bulkhead mounting shall be provided with each equipment. The junction box shall have input connectors for cables from the sonar hydrophones and the alignment signal outputs of the equipment. One-ohm terminating resistors for the balanced alignment signal outputs shall be mounted in the junction box (see figure 1). An output connector for cables to the sonar preamplifier signal input shall be provided.

3.7 Electrical requirements.-

3.7.1 Power source.- The equipment shall be designed to operate from 115 volts plus or minus 10 percent single phase a.c. at 50 c.p.s. plus or minus 5 percent, 60 c.p.s. plus or minus 5 percent and 400 c.p.s. plus or minus 10 percent.

3.7.2 Total power.- The total power required to operate the equipment shall be held to a minimum and shall not exceed 70 watts.

3.7.3 Radio interference.- The equipment shall meet the applicable radio interference requirements of MIL-I-16910. Any parts used for the reduction of radio interference, such as capacitors and filters, shall be as approved by the procuring agency.

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3.8 Mechanical requirements.-

3.8.1 Dimensions.- The dimensions of the equipment with the cover in place shall not exceed 21 inches wide by 11-1/16 inches high by 12 inches deep.

3.8.2 Weight.- The overall weight of the equipment, including the cover, shall be held to a minimum consistent with the other requirements of this specification and shall not exceed 60 pounds.

3.8.3 Finish.- The exterior finish of the equipment shall be gray enamel in accordance with type III, class 2, of MIL-E-15090.

3.8.4 Rough usage.- The design of the equipment shall be such as to preclude damage to the equipment when handled by relatively inexperienced personnel.

3.8.5 Accessibility.- The construction shall allow maximum accessibility to internal parts and subassemblies.

3.8.6 Equipment case.- The equipment shall be housed in a dripproof case in accordance with MIL-STD-108. The case and panel shall be fabricated from steel to provide an effective magnetic shield for the equipment.

3.8.6.1 Cover.- The equipment case shall include a separable hinged front cover which shall be closed when the equipment is not in use.

3.8.6.2 Accessory stowage.- A small compartment shall be provided in the equipment separable cover for stowage of the accessory and power cables and the technical manual.

3.8.6.3 Shock mounts.- The equipment cabinet shall be provided with shock mounts for mounting on a shelf or table top.

3.8.6.4 Stuffing tubes.- Stuffing tubes shall be provided on the side or back of the test set cabinet for the connection of cables to the test target hydrophone, alignment signals output, and the ship's power.

3.8.6.5 Identification plate.- The identification plates shall be affixed to the outside of the hinged front cover and to the front panel.

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3.8.7 Panel layout.- The arrangement of the equipment controls and connectors on the panel shall be functional and of maximum convenience to the operator. A drawing of the proposed panel layout shall be submitted to the bureau or agency concerned for approval prior to fabrication. See figure 2 for items required on the front panel and a possible arrangement.

3.8.7.1 Panel marking.- Panel markings shall be engraved, cast or etched into the panel. Silk screen processes shall not be used. The panel markings shall be filled with a permanent white material.

3.8.7.2 Fluorescent filling.- When required (see 6.1) fluorescent filling shall be used in the panel marking, knob pointers, meter scales, and meter pointer to facilitate operation in subdued light.

3.8.8 Operating controls.- All controls necessary for operating the equipment shall be placed on the front panel.

(a) Power switch.- A double pole, single throw switch shall be provided in the input power line.
(b) Attenuator.- A nine-position rotary switch shall be provided for selecting the number of db attenuation.
(c) Test target switch.- A momentary toggle switch spring loaded to remain in the OFF position shall be provided for actuating the test target hydrophone signal.
(d) Test-operate switch.- A five-position rotary switch shall be provided to select the alignment output function.

3.8.9 Connectors.- The following connectors shall be provided on the front panel:

(a) Meter plug.- A telephone type jack to connect external voltages to the meter circuit.
(b) AN/BQR-2B Simulated target.- A telephone jack to connect the output of the noise amplifier to external equipment.
(c) External oscillator.- A telephone type jack to connect an external signal to the alignment and metering circuits.

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3.8.10 Junction box.-

3.8.10.1 Dimensions.- The dimensions of the junction box, including the mounting flanges, shall not exceed 6-1/4 inches wide by 1-7/8 inches high by 11/16 inch deep.

3.8.10.2 Weight.- The weight of the junction box shall not exceed 1.5 pounds.

3.8.10.3 Identification plate.- The identification plate shall be affixed to the side of the junction box.

3.8.10.4 Connectors.- The input connector and the output connectors shall be mounted on the back of the junction box.

3.8.11 Indicating meter.- The indicating meter shall be a 3-1/2 inch diameter, flange mounted type and shall meet the requirements of MIL-M-10304, except as specified herein.

3.8.11.1 Accuracy.- The indicating meter shall be sufficiently accurate to allow the equipment to meet the overall requirements of this specification.

3.8.11.2 Environmental effects.- The indicating meter shall be a hermetically sealed meter to reduce the effects of heating, temperature and humidity. The effects of overload, shock, and vibration, shall be sufficiently small as to assure compliance with the other requirements of this specification.

3.8.11.3 Type.- The indicating meter shall be an a.c. rectifier type.

3.8.11.4 Color.- The indicating meter housing shall be the same color as the panel. The meter face shall be white.

3.8.11.5 Scales.- (See 3.6.6.1).

3.8.11.6 Scale lengths.- The scale lengths shall be as great as possible within the limits imposed by readability and the likelihood of operator error.

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3.9 Maintainability requirements.-

3.9.1 Plug-in construction.- Optimum use may be made of plug-in construction.

3.9.2 Interchangeability.- Mechanical and electrical interchangeability shall exist between similar assemblies, subassemblies, and replaceable parts, regardless of the manufacturer or supplier. Interchangeability does not mean identity, but requires that a substitution of such like assemblies, subassemblies, and replaceable parts be easily effected without physical or electrical modifications of any part of the equipment including cabling, wiring, and mounting. In the design of the equipment, provisions shall be made for design tolerance sufficient to accommodate various articles such as tubes, resistors, and other parts having the limiting dimensions and characteristics set forth in the specifications for the particular parts involved without departure from the specified performance.

3.10 Safety requirements.- Fuse protection and other safety protective devices shall be provided in accordance with MIL-E-16400 and as specified herein. The general design shall be planned so as to minimize the danger of shock to operating personnel.

3.10.1 Fail-safe requirements.- Operator safety shall be maintained despite failure of any parts of this equipment except failure occasioned by misapplication of high voltage exceeding the limits of this specification and causing damage to the insulation of external parts or accessories.

3.10.2 Terminals.- All terminals shall be properly spaced, insulated, and protected.

3.10.3 Power transformers.- Transformer windings used to supply rectifier filament power shall not be used to supply the filaments of other tubes.

3.10.4 Time-delay relay.- This equipment shall be so designed that it will not be damaged by turn-on voltage conditions. A time-delay relay shall not be used.

3.10.5 Control positions.- Positions of controls in any possible combination shall not damage the equipment.

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3.10.6 Fuses.- Both sides of the power line shall be fused. The power on-off switch shall be between the power line and the fuses. Spare fuses in fuse-holders shall be provided.

3.11 Value engineering requirements.- When not in conflict with the other requirements of this specification, every possible effort shall be made to minimize the cost of procuring, manufacturing, and maintaining this equipment.

3.11.1 Value engineering data.- Prior to the placement of any order for parts or materials, the contractor shall consult with representatives of the procuring agency having cognizance of value engineering and shall provide such data as may be required.

3.11.2 Specification changes.- The contractor is encouraged to suggest changes to this specification which would further the aim of this section or would improve this equipment without undesirable results.

3.12 Accessories.-

3.12.1 Voltmeter cable.- A 4-foot shielded cable for connecting external signal voltage sources to the 0 to 10 volt meter input shall be provided. This cable shall have a telephone plug at the equipment end and crocodile clips at the other end.

3.12.2 External oscillator cable.- A 4-foot shielded cable for connecting an external oscillator output to the external oscillator input of the equipment shall be provided. This cable shall have a telephone plug on the equipment end and a connector to accept the output of a military type TS-382A/U or equivalent oscillator on the other end.

3.12.3 Power cable.- The power input cable shall be an 8 foot (±3 inches) length of 3-conductor cable type CO-03LGF (3/18) 0240 in accordance with MIL-C-3432, except that the ground lead shall be green. A type UP-131M plug in accordance with MS24663 shall be attached to one end of the cable. The green conductor shall extend 4 inches beyond the end of the cable jacket and shall be attached to the grounding blade on the type UP-131M plug by the screw provided. The other end of the cable shall be attached to a type MS3106R-14S-7S connector in accordance with MIL-C-5015 and MS3106. The green wire shall be attached to pin b of the connector.

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3.12.4 Test points and test features.- Test points and test features shall be provided in accordance with MIL-STD-415.

3.12.5 Corrosion protection.- The equipment shall be protected from corrosion in accordance with MIL-E-16400.

3.12.6 Technical manuals.- In addition to any bulk quantities required by the contract, two copies of an approved technical manual prepared in accordance with MIL-M-16616 shall be provided with each equipment.

3.13 Workmanship.- Workmanship shall be in accordance with MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification.- Unless otherwise specified herein, sampling, examination, and testing of this equipment shall be in accordance with MIL-E-16400 and shall consist of the following classifications:

(a) Preproduction inspection.
(b) Acceptance inspection.

4.3 Preproduction inspection.- The complete preproduction model of the equipment shall be subjected to the preproduction inspection of MIL-E-16400, the fungi test of 4.5.1 and any other tests considered necessary by the bureau or agency concerned to determine full conformance with this specification.

4.3.1 Shipboard tests.- At the discretion of the bureau or agency concerned, the preproduction model, including installation drawings, instruction material, adequate repair parts and any other material necessary for conducting tests, shall be submitted for installation and tests aboard a naval ship. These tests will include such operational tests under service conditions as may be considered necessary to prove full conformance with this specification.

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4.3.1.1 Operating test.- The shipboard tests of 4.3.1 shall specifically include a trial with at least one sonar receiving equipment, such as AN/BQR-3, in which satisfactory performance as a test and alignment set shall be demonstrated.

4.3.2 Disposition.- The preproduction model shall be modified or corrected as shall have been shown to be required as a result of the preproduction inspection. Upon incorporation of all the changes shown to be required by the preproduction inspection into the design of the production equipment, the equipment will be released for production.

4.4 Acceptance inspection.- Acceptance inspection shall consist of the production inspection of 4.4.1 and the production control inspection of 4.4.2.

4.4.1 Production inspection.- The production inspection of MIL-E-16400 shall be performed on each equipment.

4.4.2 Production control inspection.- Production control inspection shall be conducted on one equipment of every 25 equipments, and shall consist of the production control inspection of MIL-E-16400.

4.5 Test procedure.-

4.5.1 Fungi test.- The parts specified in 3.3.2.1 shall be subjected to the fungi test of MIL-T-945.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Basic equipment.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

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5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.- Onboard repair parts shall be preserved and packaged Level A; packed Level C and marked Levels A and C in accordance with Specification MIL-E-17555. The Level A packaging method shall be modified in accordance with Standard MIL-STD-758.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1).

5.2.1.1 Preservation, packaging, packing and marking.- The equipment accessories, technical publications and repair parts shall be preserved and packaged by level A or C; packed by level A or B and marked, as specified in the contract or order in accordance with MIL-E-17555. Method II shall apply for level A packaging of equipment except that Method IC5 shall apply where the storage time before use will not exceed 6 months. For submarine repair parts, the level A packaging method shall be modified in accordance with MIL-STD-758.)

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) If fluorescent filling is required in the panel marking knob pointers, meter scales and meter pointer (see 3.8.7.2).
(c) Quantity of technical manuals.
(d) Preservation, packaging, packing and marking requirements if other than those required by 5.1 (see 5.2).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 6625-N236Sh)

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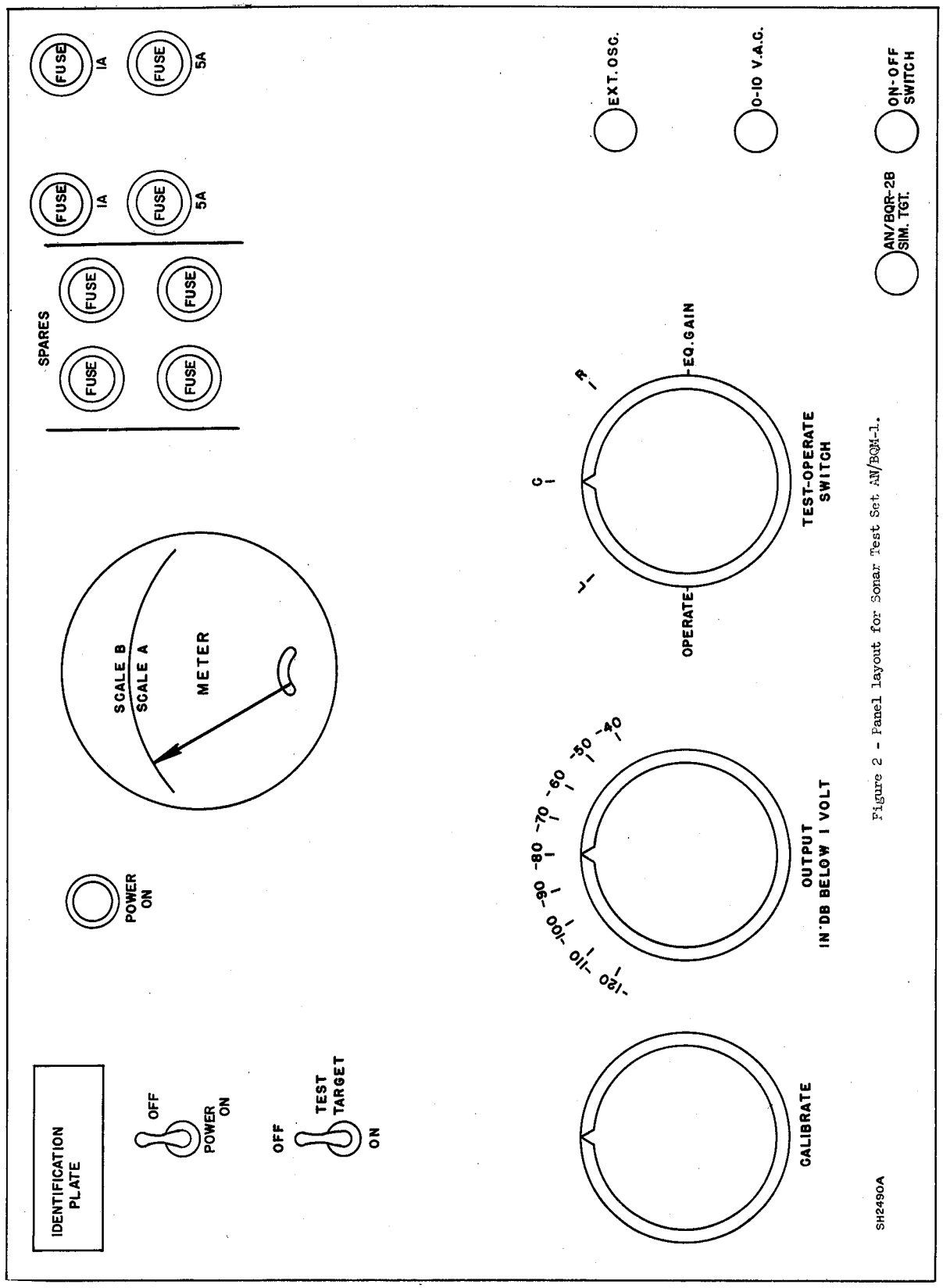


Figure 2 - Panel layout for Sonar Test Set AN/BQM-1.

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